BioHackrXiv

BioHackathon series:
Biohackathon SWAT4HCLS 2026
AmsterdamUMC
OpenMRS

Submitted: 09 Apr 2026

SWAT4HCLS26 report: OpenMRS as a reference implementation platform for Electronic Health Records and Clinical Decision Support

Andra Waagmeester [1], Stephanie "Ace" Medlock [1,2], Eric Herman [2,3], Claude Nanjo [4], Chang Sun [5], and Ronald Cornet [1]

1 Department of Medical Informatics, AmsterdamUMC, Location AMC Amsterdam, the Netherlands **2** Stichting OpenElectronicsLab, The Netherlands **3** Commons Caretakers B.V., Amsterdam, The Netherlands **4** University of Utah **1** DACS - Data Department of Advanced Computing Sciences, Faculty of Science and Engineering, Maastricht, the Netherlands

Introduction

Electronic health records (EHRs) have become the central piece of software in modern health care. [Shen2025] Most patient care is documented and managed in the EHR, and consequently, it has long been understood that functionality to support clinical decisions should also be integrated into the EHR. [Kawamoto2005] However, shareability of clinical decision support systems (CDSSs) is a longstanding challenge, partly due to limited interoperability of EHRs. [Sittig2007, Shen2025] This makes it difficult for researchers to show how a CDSS - especially a novel CDSS - could be integrated into an EHR. Innovations in CDSSs can be disseminiated by publishing a specification, but a reference implementation for the CDSS would help potential users imagine how the CDSS could be integrated into their workflow and support them in practice, and help technical implementers understand what is required to fully implement the specification as an EHR-integrated CDSS.

Open-source EHRs offer a unique opportunity for CDSS researchers to create a reference implementation that demonstrates their vision on how their CDSS can be integrated with an EHR and the clinial workflow. The open-source licensing model gives users the freedom to adapt the software to their needs, [gnu.org] from simply using or adding functionality that implements existing interoperability standards, to adding new functionality that demonstrates novel means of deeply integrating new kinds of support.

Additionally, open-source EHRs offer unique opportunities for educating medical informatics professionals. EHRs are central to modern health care, but many medical informatics students never get hands-on experience with an EHR until after graduation. The open-source licensing model also gives users the "freedom to inspect," [gnu.org] meaning that students can "look under the hood" of an EHR, as well as gain hands-on experience with various secondary uses of the EHR, including uses which require interoperability.

Due to the authors' previous experience with OpenMRS, [Medlock2022] OpenMRS was chosen as the target system for our Biohackathon table. The broad goal for the OpenMRS table was: What steps are needed to use OpenMRS as a resource for reference implementations?

This can be divided into 4 levels:

1. Use OpenMRS as a data source, using its available APIs A copy of the reference implementation for OpenMRS [OpenMRS] was created, with demo data, and provided for use during the BioHackathon. The goal for this level is simply to see if a demo application for any desired functionality can be implemented simply by using the APIs that are already preset in the OpenMRS reference implementation. (Note: The OpenMRS

reference implementation should not be confused with a reference implentation for new CDSS functionality. The OpenMRS reference implementation shows how OpenMRS can be implemented; our goal is to implement a reference implementation for novel CDSS functinality *integrated with* OpenMRS.)

2. Use OpenMRS running locally, and directly access its database The OpenMRS community provides instructions for installing OpenMRS on a local computer. [OpenMRS2] The goal for this level is to follow the provided instructions, create more detailed instructions if needed, and then build a demo application that queries the OpenMRS database. This can be used for applications that need to use the data from an EHR, but do not need to interact with users via the EHR interface.
3. Add a module to OpenMRS This is the first step toward making modifications to OpenMRS itself. The OpenMRS community provides some "instructions for developers". [OpenMRS3] The goal for this level is to follow those instructions (and improve them where needed) with the goal of adding a simple module (e.g. "add a clickable URL to the patient summary screen"). This is what is needed to embed new functionality in OpenMRS.
4. Build OpenMRS from source To make deep changes to OpenMRS, rather than just adding something to it, new functionality may need to be added to existing OpenMRS modules. The goal of this level is to produce instructions for building a minimum viable installation of OpenMRS from source.

An additional goal of this table is to produce a "roadmap" for EHR education, describing what medical informatics students should learn about an EHR and how an open source EHR could be leveraged to teach it.

Open-Source EHR as a reference implementation

- Level 1 (OpenMRS APIs): We noted that the version of FHIR implemented in the OpenMRS reference implementation is quite old, thus contributing to the maintainence of the FHIR module may be a good long-term goal. [OpenMRS_FHIR2]
- Level 2: (OpenMRS local installation): See notes below
- Level 3 (OpenMRS development): https://gitlab.com/openmrs_education/openmrs-sdk-sandbox
- Level 4 (OpenMRS from source): https://gitlab.com/ericherman/openmrs-sandbox

Open-Source EHR in Medical Informatics Education

The figure below describes a long-term vision for using an open-source EHR in medical informatics education. Prerequisites are that students can install the software (and any supporting software that's needed) without needing deep knowledge of system administration. The use cases describe a mix of education and research goals (which have considerable overlap in our education program, since it includes education about research methods).

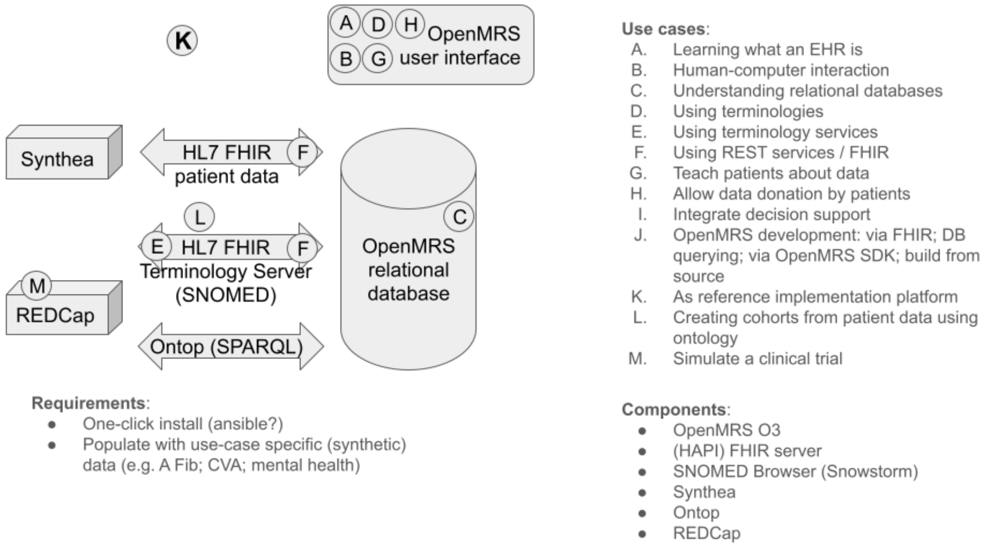

Figure 1: Long term vision

Sources:

- Shen Y, Yu J, Zhou J, Hu G. Twenty-Five Years of Evolution and Hurdles in Electronic Health Records and Interoperability in Medical Research: Comprehensive Review. J Med Internet Res. 2025 Jan 9;27:e59024. doi: 10.2196/59024.
- Kawamoto K, Houlihan CA, Balas EA, Lobach DF. Improving clinical practice using clinical decision support systems: a systematic review of trials to identify features critical to success. BMJ. 2005 Apr 2;330(7494):765. doi: 10.1136/bmj.38398.500764.8F.
- Sittig DF, Wright A, Osheroff JA, Middleton B, Teich JM, Ash JS, Campbell E, Bates DW. Grand challenges in clinical decision support. J Biomed Inform. 2008 Apr;41(2):387-92. doi: 10.1016/j.jbi.2007.09.003.
- https://www.gnu.org/philosophy/free-sw.en.html accessed 2026-04-09
- Medlock S, Ploegmakers KJ, Cornet R, Pang KW. Use of an open-source electronic health record to establish a "virtual hospital": A tale of two curricula. Int J Med Inform. 2023 Jan;169:104907. doi: 10.1016/j.ijmedinf.2022.104907.
- https://github.com/openmrs/openmrs-distro-referenceapplication accessed 2026-04-09
- https://openmrs.atlassian.net/wiki/spaces/docs/pages/150930190/Set+Up+an+Instance+of+O3 accessed 2026-04-09
- https://openmrs.atlassian.net/wiki/spaces/docs/pages/25477022/Getting+Started+as+a+Developer accessed 2026-04-09
- https://github.com/openmrs/openmrs-module-fhir2 accessed 2026-04-09

Install reference implementation

Linux

To install on Linux: `apt update` `apt install docker-compose-v2` `git clone https://github.com/openmrs/openmrs-distro-referenceapplication.git` `cd openmrs-distro-referenceapplication` `git tag --sort=-v:refname` `TAG=3.5.0` `docker compose -f docker-compose.yml up -d` The tag number is replaced with the latest tag number (or whichever you choose from the list produced by the previous command). After a few minutes, the installation UI can be accessed at http://localhost:8080/openmrs/ . The initial configuration can be done here. After 15-45 minutes (depending on the speed of the underlying machine) the O3 UI can be accessed at http://localhost:8080/openmrs/spa/home .

The default login is username: admin, password: Admin123 .

Windows

```
gh repo clone openmrs/openmrs-distro-referenceapplication
cd openmrs-distro-referenceapplication
docker compose up (or docker compose up -d)
```

Waited — saw empty screen / 504s for ~30 min (first-time data load)

```
docker compose restart backend (when stuck)
```

Waited ~17 min for backend to finish loading Accessed http://localhost/openmrs/spa, logged in with admin / Admin123

Initial screen after succesful reference implementation

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRediT). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from CRediT. An example looks like this:

```
authors:
  - name: First Author
    affiliation: 1
    orcid: 0000-0000-0000-0000
    role: Conceptualization, Writing - review & editing
```

A full examples

A full example then has this structure:

```
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Formatting

This document use Markdown and you can look at this tutorial.

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

```
Table: Note that table caption is automatically numbered and should be
given before the table itself.

| Header 1 | Header 2 |
| -------- | -------- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:

Figure 2: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethodIn:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- citesAsDataSource: when you point the reader to a source of data which may explain a claim
- usesDataFrom: when you reuse somehow (and elaborate on) the data in the cited entity
- usesMethodIn
- citesAsAuthority
- citesAsEvidence
- citesAsPotentialSolution
- citesAsRecommendedReading
- citesAsRelated
- citesAsSourceDocument
- citesForInformation
- confirms
- documents
- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

...

Acknowledgements

...

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**

Author contributions

Andra Waagmeester: Writing – original draft Stephanie "Ace" Medlock: Writing – original draft Eric Herman: Conceptualization Claude Nanjo: Writing – original draft Chang

Sun: Conceptualization Ronald Cornet: Writing – original draft